

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 04 — INTRODUCTION TO SUPERVISORY CONTROL AND DATA ACQUISITION (SCADA)

Student Edition

Authorized ICDFA training environments only
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Lab 04: Introduction to Supervisory Control and Data Acquisition (SCADA)

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 3–4 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Explain the purpose and architecture of SCADA systems.
- Identify the roles of PLCs/RTUs, HMI, alarms, event recording and historization.
- Start FUXA in the training SCADA VM and load a prepared HMI project.
- Connect a SCADA HMI to the OpenPLC Runtime over Modbus TCP in the isolated lab.
- Observe an automated tank-fill process and interpret operator-facing indications.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	172.16.100.137	plc	password
SCADA VM	172.16.100.141	scada	password

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

LAB 04: FuXa and OpenPLC Simulation

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Course: ICDFA Advanced Critical Infrastructure Security

Lab Number: LAB 04

Date: 27 September 2026

Objective

To integrate OpenPLC Runtime with FuXa SCADA using Modbus TCP communication and observe a tank-fill process through an HMI interface.

Procedure Summary

Step 1

Started the ICDFA PLC VM and ICDFA SCADA VM using the isolated host-only network.

Step 2

Logged into the SCADA VM and started FuXa.

Step 3

Opened the FuXa editor at:

<http://127.0.0.1:1881/editor>

Step 4

Loaded the provided project:

TankFillTest.json

Step 5

Modified the OpenPLC connection settings in FuXa:

Protocol : Modbus TCP

Endpoint : 172.16.100.137:502

Step 6

Logged into OpenPLC Runtime.

Step 7

Uploaded: TankFillSim.st

Program Name: Tank Fill Sim

Step 8

Started OpenPLC Runtime and confirmed:

Status: Running

Step 9

Returned to FuXa and launched the Tank Fill HMI.

Step 10

Observed process behavior through the HMI dashboard.

Observations

Tank Level

The tank level was displayed dynamically on the graphical tank visualization.

Valve Status

The fill valve opens when the PLC requests filling and closes when the tank reaches the defined level.

Communication

The HMI communicated with OpenPLC through:

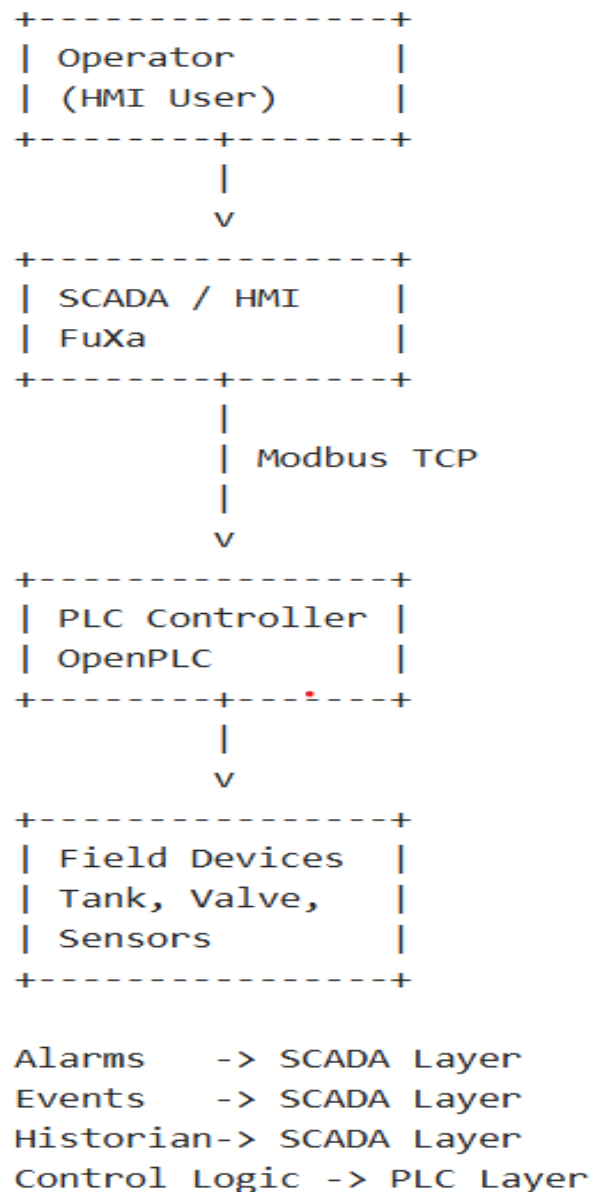
Modbus TCP

Data Visualization

The operator interface provided:

- Real-time tank level
- Tank fill status
- Valve indication
- Trend chart area
- PLC communication status

SCADA Architecture Diagram



PLC vs SCADA Explanation

A PLC and a SCADA system perform different but complementary functions within an industrial control system. The PLC is responsible for executing real-time control logic. It continuously reads inputs from sensors, processes the control program, and writes outputs to field devices such as valves, motors, and pumps. In this laboratory exercise, OpenPLC executed the tank-fill control algorithm, determining when the fill valve should open or close based on the tank level conditions.

The SCADA/HMI system operates at a supervisory level. Rather than directly controlling the process, it provides visualization, monitoring, alarming, reporting, and operator interaction capabilities. FuXa displayed the tank, level indicator, valve status, and process information while receiving data through Modbus TCP from OpenPLC. The SCADA interface allows operators to understand the state of the process quickly and respond to abnormal conditions.

The PLC focuses on deterministic control and reliability, whereas SCADA focuses on situational awareness and decision support. Together they create a complete industrial automation solution in which the PLC controls the process and the SCADA system presents information, records historical data, manages alarms, and allows operators to supervise the overall operation safely and effectively.

Knowledge Check

1. What is the primary purpose of an HMI in a SCADA system?

Response:

The primary purpose of an HMI is to provide a graphical interface that allows operators to monitor process conditions, view equipment status, acknowledge alarms, and interact with industrial control systems.

2. What is the difference between an alarm, an event record and historized process data?

Response:

An alarm indicates an abnormal condition that requires attention. An event record logs significant actions or system changes such as start, stop, or operator commands. Historized process data consists of time-stamped measurements collected over time for analysis, reporting, and trending.

3. Why should a SCADA operator have context rather than only raw sensor values?

Response:

Context helps operators understand the meaning of sensor values by displaying trends, equipment status, alarms, and process relationships. This allows faster and more accurate operational decisions.

4. What role does the PLC play compared with the SCADA application?

Response:

The PLC performs the real-time control of industrial processes by executing logic and controlling devices, while the SCADA application supervises the process through monitoring, visualization, alarm management, data collection, and operator interaction.

Evidence available in Screenshots folder

1. OpenPLC Runtime Running Screenshot
2. FuXa Tank Fill HMI Screenshot
3. SCADA Architecture Diagram
4. PLC vs SCADA Explanation